IMMUNE

Autonomous Infrastructure Immune System

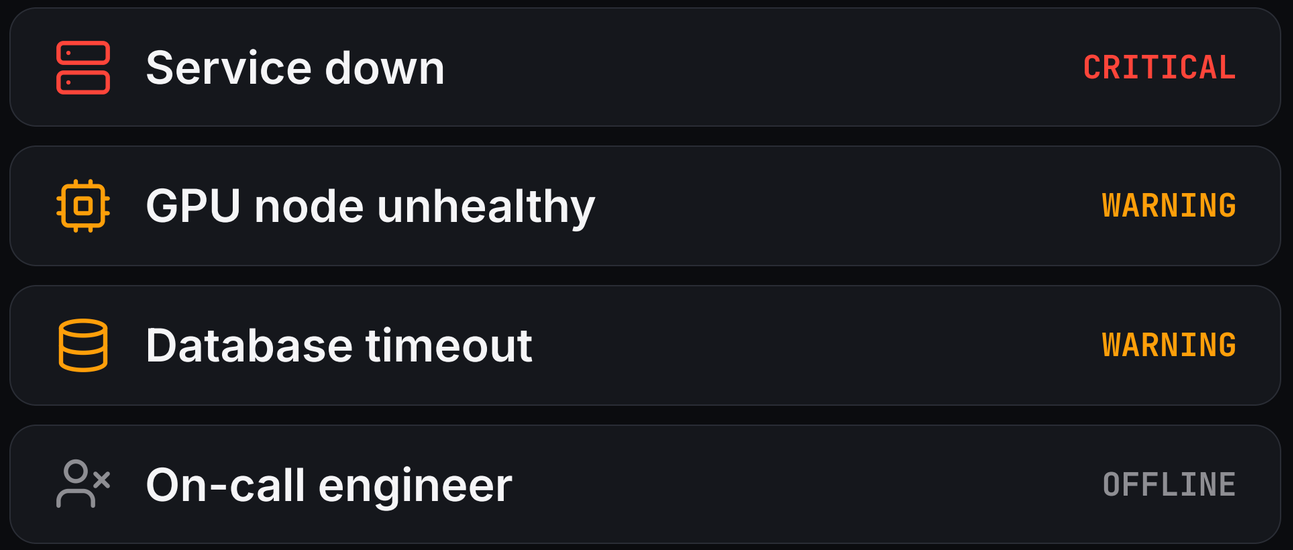


讓系統學會自救。

4 ALERTS · 0 RESPONDERS

03:17

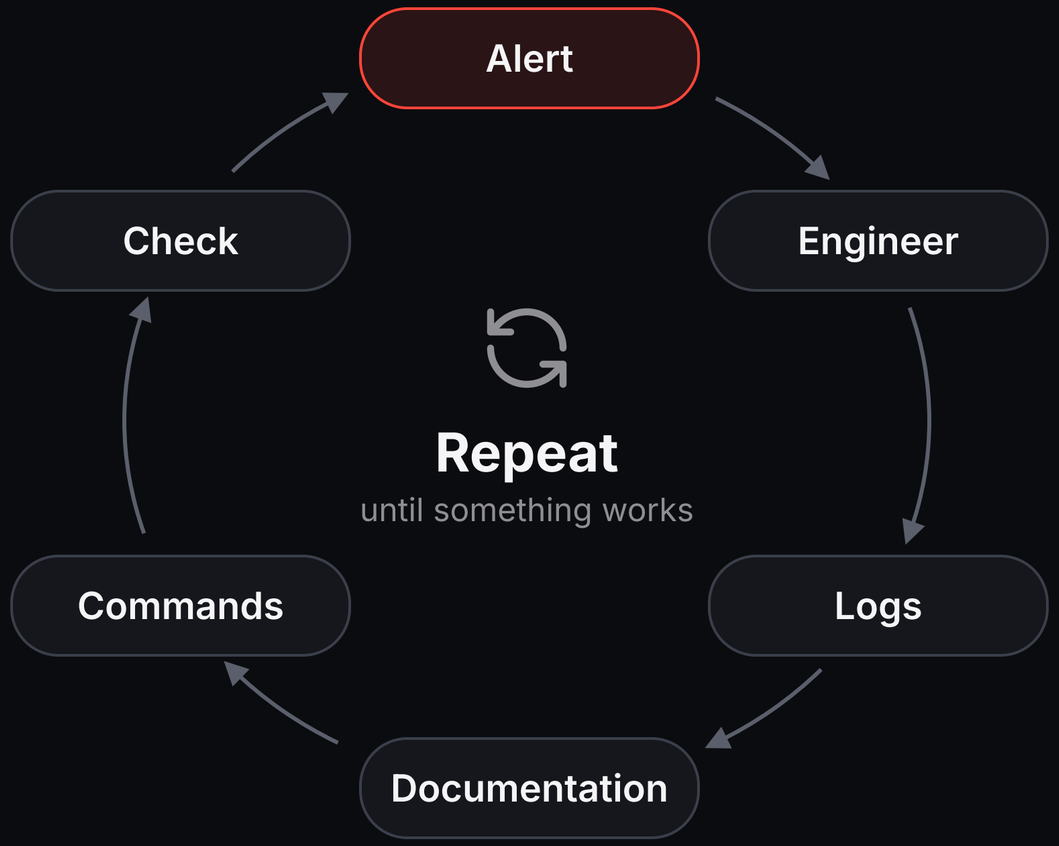
AM. Everything is paging.



Systems can detect failure.

But they still cannot recover themselves.

How incidents are handled today



Every incident becomes a human debugging loop.

Read the alert, wake someone up, dig through logs and docs, try commands, check, and go around again.

What if infrastructure had an immune system?

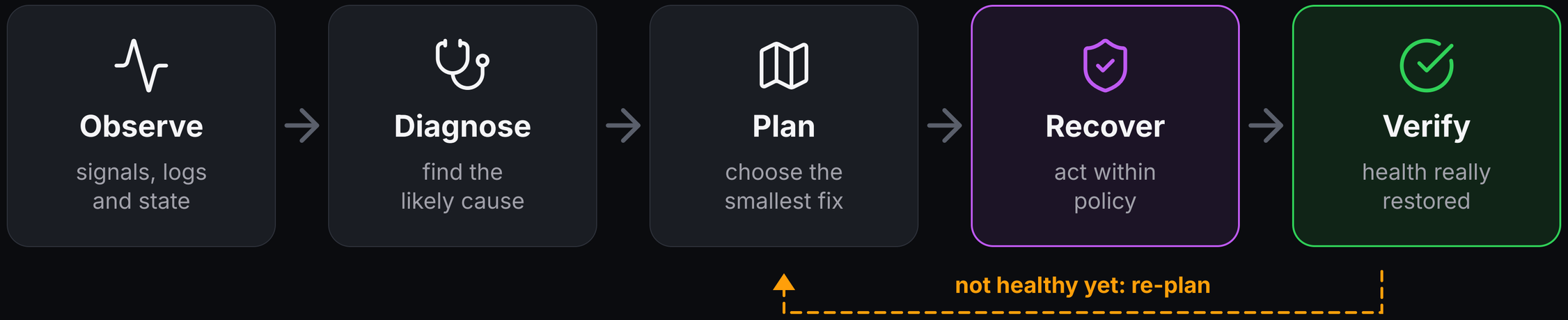


LOCAL-FIRST · SELF-HEALING · POLICY-GOVERNED

Meet IMMUNE.

Observe. Diagnose. Recover. Verify.

A closed-loop recovery agent.



Knowledge runbooks and past recoveries

Policy what may run, what needs a human

Local AI a reasoning model running on-site, on replaceable hardware

Monitoring tells you something is wrong.
IMMUNE tries to make it right.

TRADITIONAL MONITORING

Detect → **Alert a human** → **STOP** *the rest is manual*

IMMUNE

Detect	→	**Understand**	→	**Respond**	→	**Recover**	→	**Learn**
sensing		*recognition*		*immune response*		*verified recovery*		*immune memory*

Self-healing. Not self-destructive.

AI proposal the model suggests a fix

↓

Policy is this allowed, here and now?

↓

Allowed action a named action, never a shell

↓

Execute the smallest change that fixes it

↓

Verify status and content must pass

Blocked actions
destructive or out-of-policy commands never run

Audit log
every decision and action, replayable

Optional human approval
risky steps wait for a person

PROTOTYPE SAFETY TEST
3/3 injected attacks blocked · 0 destructive actions

Intelligence stays where the failure happens.

TRADITIONAL AI OPERATIONS



depends on cloud connectivity

IMMUNE



works without WAN or cloud connectivity

LOCAL

model, knowledge and actions on-site

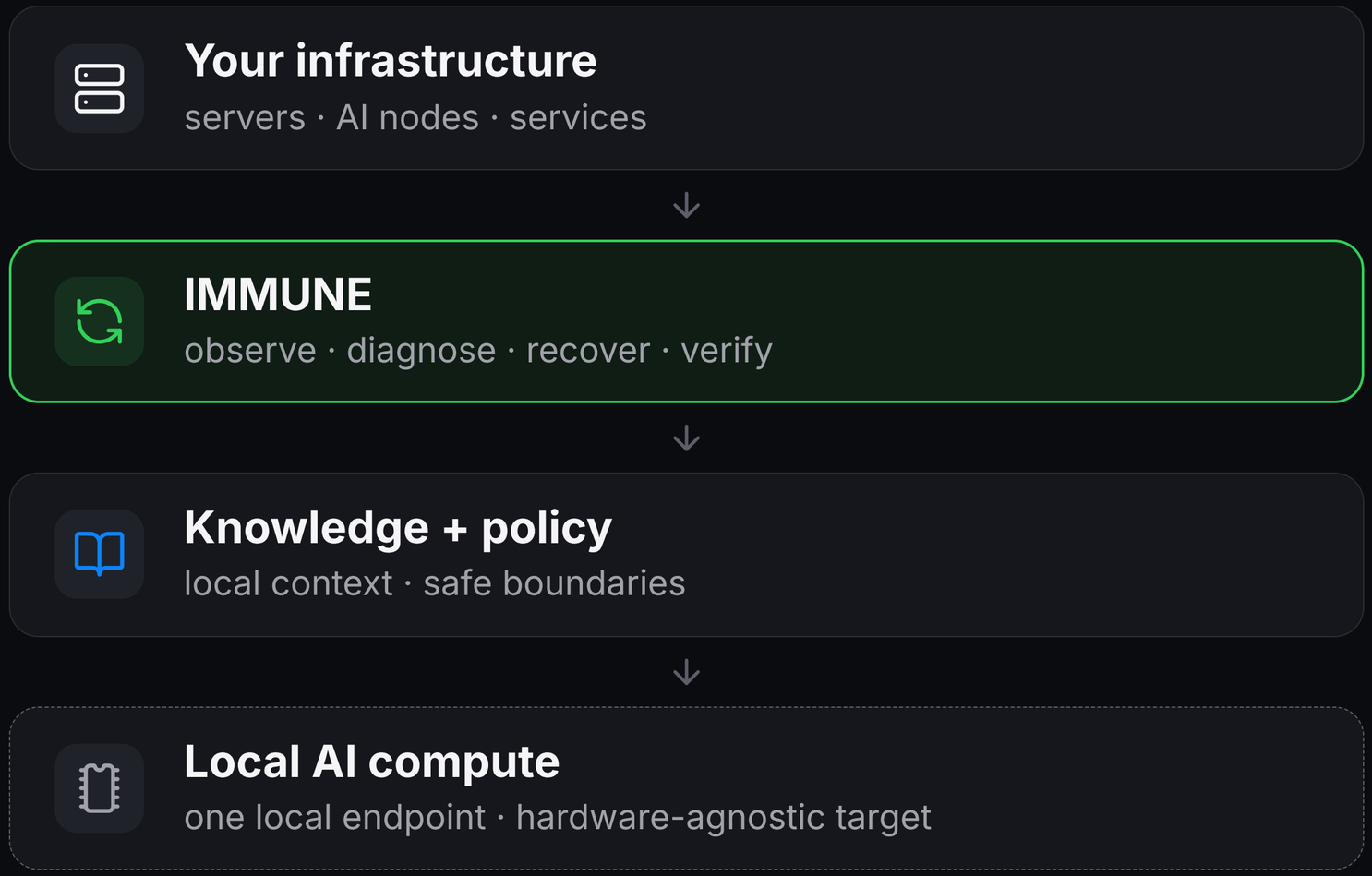
PRIVATE

logs and data never leave

AVAILABLE

recovery works when the WAN is down

Four layers. One closed loop.



AUDIT TRAIL

Every decision is recorded.

signals, approvals and actions, replayable

How IMMUNE is built.

SOLID = BUILT TODAY · DASHED = PLANNED



1 signals
2 incident with evidence
3 reason and retrieve
4 proposed action
5 allowed action
6 verified: close · failed: re-plan
7 verified fix becomes memory
planned for the competition

More than an idea.

50
fault scenarios designed, in 8 failure classes

27
test runs on the prototype, covering 15 of them

23
recovered with no human: 23 of 27 runs (85%)

0
runs made worse, and 0 false “healthy” reports

Local agent
reasons on-site

Fault injection
50 fault scenarios

Knowledge retrieval
runbooks and past fixes

Tool execution
allow-listed actions

Recovery loop
act, check, try again

Verification
status and page content

Re-planning
when a fix does not hold

Audit trail
every step, replayable

Prototype device, local test environment with containerized web services. Each of the 15 fault types run was recovered at least once. The method and results are documented in the repository.

From prototype to infrastructure immune system.

TODAY · BUILT

Agent recovery prototype

- ✓ Local recovery agent
- ✓ Fault-injection test bench
- ✓ Safety gates and audit trail
- ✓ Measured recovery results

grows into

COMPETITION PLAN

Infrastructure immune system

- Hardware-agnostic runtime
- Expanded failure scenarios
- Stronger safety policies
- Evaluation framework
- Production-oriented UX

How we will prove it works.

Faster

MTTR ↓

time from failure to verified recovery

More autonomous

recoveries without a human ↑

share of incidents closed with no intervention

Safer

unsafe actions → 0

no destructive or out-of-policy action runs

More reliable

verified recovery ↑

status and content checked, no false “healthy”

Competition evaluation will compare IMMUNE and manual operations on identical injected failures.

`CURRENT PROTOTYPE` 23 of 27 runs recovered with no human · no manual-ops comparison yet

Anywhere downtime matters.



AI infrastructure

GPU, inference and training nodes



Enterprise systems

servers, services and containers



Remote and edge systems

systems with no guaranteed human on site

From monitoring to self-recovery.

TODAY

Infrastructure can detect failure.

IMMUNE

It can observe, diagnose, recover and verify, locally.

NEXT

Turn the working prototype into a hardware-agnostic infrastructure immune system.

Less downtime. Less human intervention. Safer autonomous recovery.

IMMUNE

Infrastructure should not just detect failure.

It should know how to recover.

OBSERVE. DIAGNOSE. RECOVER. VERIFY.